RETAILPULSE

CASE STUDY / NO. 001

E-COMMERCE CUSTOMER & REVENUE INTELLIGENCE

RetailPulse

A data intelligence teardown of a UK online retailer — reading the heartbeat of 541,909 transactions to diagnose churn, reveal buying patterns, and prescribe a path back to healthy growth.

SQL · Python · PostgreSQL · RFM Segmentation · Cohort Analysis · Market Basket Analysis · Tableau

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Source dataset
UCI Online Retail · Dec 2010 – Dec 2011

EXECUTIVE SUMMARY

The business at a glance

A medium-sized UK online retailer — home decor, accessories and personalised gifts, £7–10M in annual revenue — came in with a familiar symptom: healthy top-line numbers masking a customer base that was quietly churning. Twelve months of invoice-level transactions were rebuilt into a governed data model and read through three lenses — **RFM segmentation**, **cohort retention** and **market basket analysis** — to turn 541,909 rows of raw transactions into a prioritised action plan.

4,373
Unique customers

25,733
Orders received

£10.7M
Total revenue

5,172
Cancelled orders (20%)

£894K
Revenue lost to cancellations

4,052
Distinct products sold

36
Countries served

20
Avg. items per basket

RISK

27.1% of customers are at risk of churn

The single largest RFM segment — 1,176 buyers who have measurably slowed down and are one bad experience from leaving for good.

OPPORTUNITY

Two products anchor 4 of the top 5 baskets

Stock code 85099B co-occurs with three other bestsellers at 600+ purchases each — a ready-made bundling and cross-sell lever.

WINDOW

Retention is decided in the first 60 days

Cohort tracking shows most customer groups lose over half their base within 2–3 months of the first purchase.

What's inside

01 · Foundation

Business context, methodology, the data model and the cleaning pass that made 541,909 raw rows analysis-ready.

02 · Diagnosis

Trend, geography, RFM segmentation, cohort retention and market basket analysis — each finding tied to a specific number.

03 · Action

A prioritised playbook and 90-day roadmap translating every finding into a concrete next step.

BUSINESS CONTEXT

Reading the symptoms before the diagnosis

CLIENT PROFILE

Mid-sized UK online retailer

Sells home decor, accessories and personalised gifts across the UK and neighbouring markets. Estimated annual revenue of £7–10M, built on a loyal but under-analysed customer base and an e-commerce platform generating thousands of transactions a day.

£7–10M
Annual revenue

36
Markets served

12 mo
Transaction history analysed

Four problems compounding each other

01

Limited customer insight

No structured view of who the most valuable customers were, or what they bought — marketing and retention spend went out untargeted.

02

Inefficient bundling & cross-sell

Product affinities were invisible without market basket analysis, leaving bundling, discounting and basket-value opportunities on the table.

03

Retention & churn

A large share of customers bought once or twice and disappeared, pushing customer acquisition cost up as effort chased new buyers instead of existing ones.

04

Data quality & structure

The raw export was inconsistent, duplicated and full of gaps — unusable for reliable analysis without a rigorous cleaning pass.

- Missed revenue from ineffective bundling strategies
- High churn stagnating customer lifetime value
- Inefficient inventory from unclear demand signals

METHODOLOGY

A five-stage path from raw export to insight

Every stage was built to be reproducible in SQL and Python, so the pipeline — not just the findings — could be handed back to the client's team.

1 Data migration
541,909 raw invoice-item rows loaded into a Postgres staging table.

→

2 Preprocessing
Nulls, duplicates, negative quantities and zero prices resolved systematically.

→

3 Normalisation
Flat export decomposed into a four-table relational schema.

→

4 EDA
Trend, geography and basket-size patterns surfaced before modelling.

→

5 Advanced analytics
RFM, cohort retention and market basket analysis layered on top.

Why these three techniques

R·F·M

RFM Segmentation

Scores every customer on how recently, how often and how much they buy — turning 4,373 individuals into five actionable groups.

Δt

Cohort Analysis

Groups customers by acquisition month and tracks survival over time to expose exactly when — not just whether — churn happens.

A→B

Market Basket Analysis

Mines co-purchase patterns across invoices to find products that sell each other.

Tooling

- PostgreSQL — staging, cleaning & the star-style analytical schema
- Python — RFM scoring, cohort matrices & basket-pair mining
- Tableau — trend, geography & segment visualisation for stakeholders

DATA FOUNDATION

From one flat file to a governed schema

The raw CSV carried invoice, product and customer attributes in a single wide table. It was decomposed into four normalised entities to eliminate redundancy and support fast, reliable querying.

A bridge table (`invoice_items`) preserves a clean many-to-many relationship between invoices and products while `is_cancelled` — derived from the "C"-prefixed invoice IDs — lets every downstream query filter cancellations without touching raw text.

Source columns

Column	Description	Type
invoice_no	Unique transaction identifier	String
stock_code	Unique product identifier	String
description	Product name / detail	String
quantity	Units purchased in the transaction	Integer
invoice_date	Date and time of the transaction	DateTime
unit_price	Price per unit	Float
customer_id	Unique customer identifier	Integer
country	Country of purchase	String

DATA QUALITY

Cleaning the record before trusting it

A rigorous pass through nulls, duplicates and structural inconsistencies — every action logged so the transformation stays auditable.

135,080

null Customer IDs

Traced to manual, in-store style entries. Imputed with a `'00000'` placeholder to preserve row-level integrity rather than dropping revenue-bearing transactions.

4,900

duplicate rows removed

Exact duplicate invoice-item records identified and dropped to prevent them from skewing frequency and revenue calculations.

10,587

negative-quantity transactions

Verified against the "C"-prefixed invoice convention as cancellations, and encoded into a dedicated `is_cancelled` flag instead of being discarded.

2,516

zero unit-price rows

1,180 non-cancelled zero-price rows investigated; 1,003 recovered via mean-price imputation by stock code, 38 dropped as unrecoverable entry errors.

Cleaning log

01 Checked nulls — 1,454 missing descriptions, 135,080 missing customer IDs.

02 Removed 4,900 duplicate transaction rows.

03 Flagged 10,587 negative-quantity rows as cancellations via `is_cancelled` .

04 Resolved 2,516 zero-price rows — imputed 1,003, dropped 38 unrecoverable entries.

05 Removed rows with irrelevant descriptions ("amazon", "check", "dotcom", etc.).

06 Imputed missing descriptions with a standard placeholder — zero nulls remaining.

07 Reviewed 2,770 non-standard stock codes — retained, as all mapped to valid transaction types.

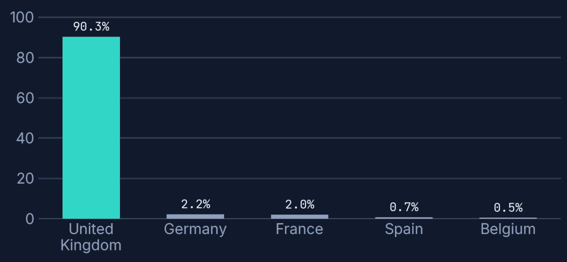
✓ Final validation — clean, structured, ready for advanced analytics.

EXPLORATORY ANALYSIS

A UK-anchored business with a long international tail

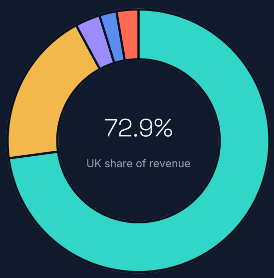
Before modelling, the shape of the business itself needed to be understood — where customers sit, and how much of the revenue they actually represent.

Customers by country share of 4,373 total



90.3% of customers are UK-based — the long tail of 35 other countries contributes the remaining 9.7%.

Revenue by country top contributors



- United Kingdom — 72.9%
- Germany — 19.2%
- Netherlands — 3.1%
- France — 2.1%
- EIRE — 2.7%

Reading the gap

Germany holds just 2.2% of customers but drives 19.2% of revenue — a small, high-value international segment worth a dedicated retention play rather than being folded into a generic "rest of world" bucket. On average, transactions carry 20 items, with a single order peaking at 1,114 units — evidence of a wholesale/bulk-buyer tail sitting inside an otherwise consumer dataset.

Basket-size distribution

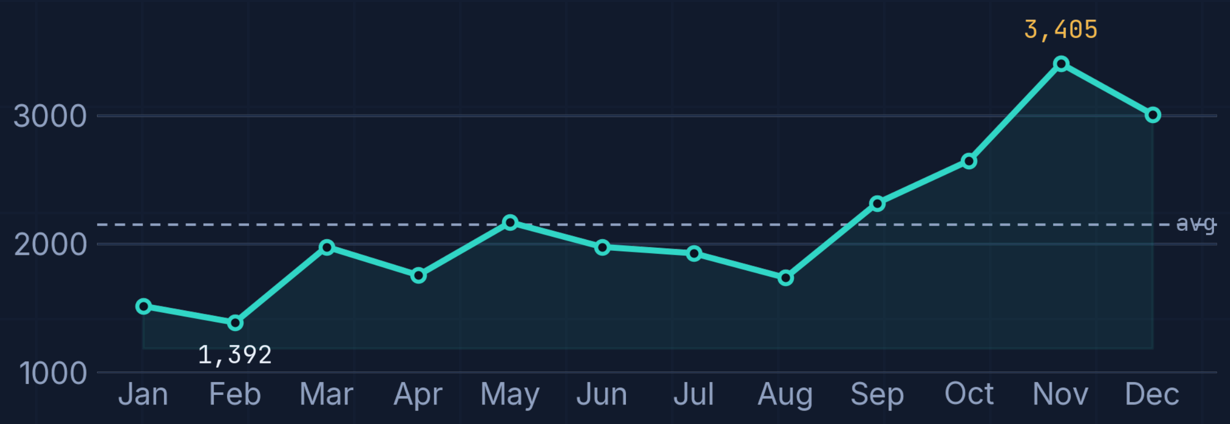
20
Average items per basket

1,114
Largest single transaction (units)

4,052
Distinct SKUs in catalogue

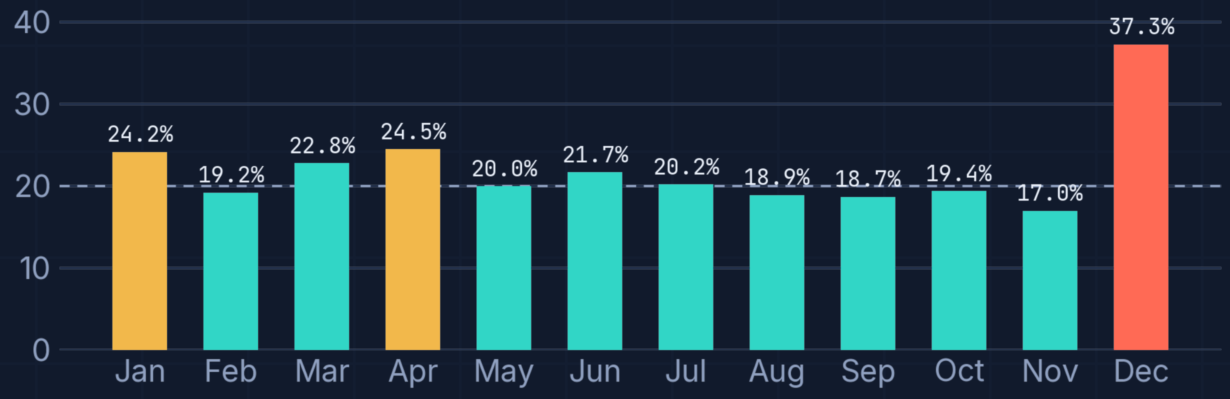
Orders climb into Q4 — cancellations spike right behind them

Unique orders per month Dec 2010 – Dec 2011



February is the trough (1,392 orders); November peaks at 3,405 — a 145% swing across the year, consistent with Black Friday and pre-holiday demand.

Order cancellation rate % of monthly orders



Cancellations hover near a 20% baseline for most of the year, but **December spikes to 37.3%** — nearly double any other month — even with data truncated at December 9th. November, by contrast, holds the lowest rate of the year (17.0%).

SEASONALITY

A 145% swing, Feb to Nov

Order volume more than doubles from the winter trough to the Black Friday peak — the clearest seasonal signal in the dataset.

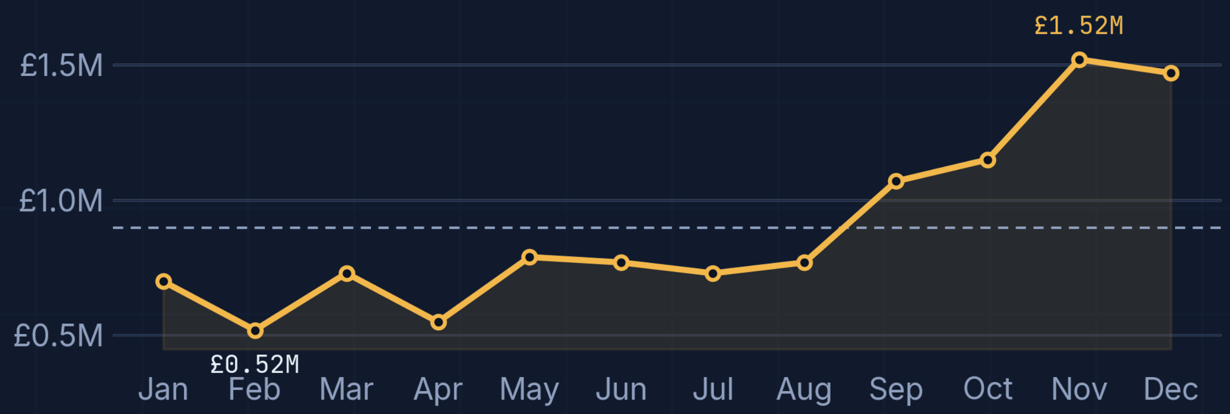
Q1 PATTERN

Post-holiday cancellations run hot

January and April both cross 24% — likely returns and order changes following the prior peak season.

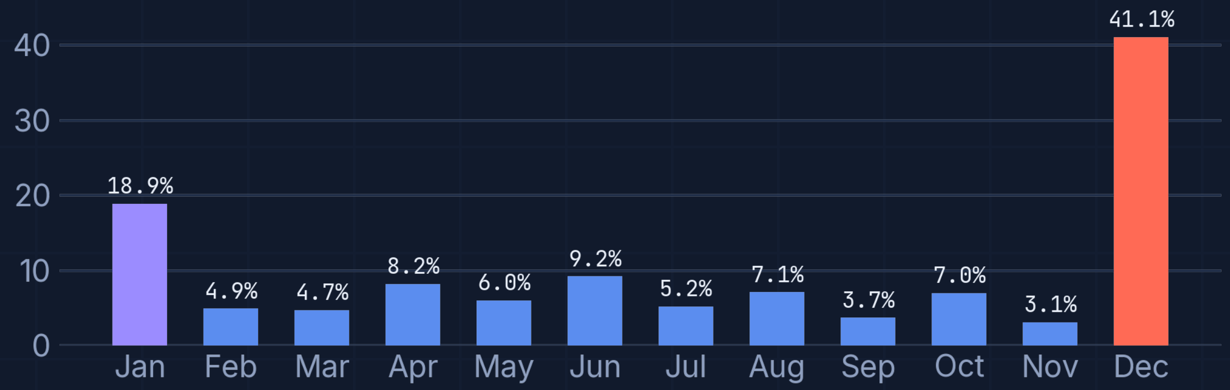
Revenue tracks orders closely — until December breaks the pattern

Total revenue per month £ millions



Revenue mirrors order volume almost exactly, climbing from a £0.52M February low to a £1.52M November peak — the clearest single indicator that demand, not pricing, drives this business.

Revenue lost to cancellations % of monthly revenue



November converts its record volume into revenue almost untouched — just 3.1% lost. December erases that discipline: **41.1% of December's revenue is lost** to cancellations, returns and failed deliveries, even on a partial month.

DECEMBER ALERT

The single clearest signal in the dataset: order volume stays strong into December, but nearly 4 in 10 of those pounds never convert to kept revenue. This is a post-purchase and returns-policy problem, not a demand problem — and it is fully addressable.

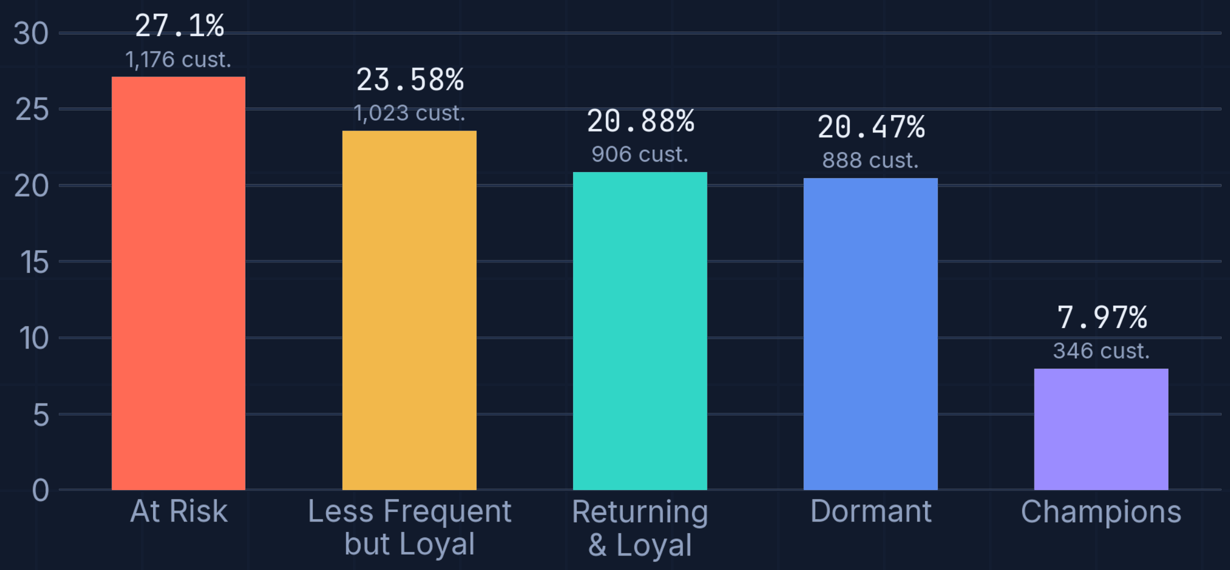
CUSTOMER SEGMENTATION

RFM turns 4,373 customers into five decisions

Each customer is scored 1–5 on **Recency**, **Frequency** and **Monetary value**; the combined score defines the segment.

Segment	Criteria	Definition
Champions	R=5, F=5, M=5	Most engaged, frequent, high spenders
Returning & Loyal	RFM ≥ 12	Consistent repeat buyers
Less Frequent but Loyal	RFM ≥ 9	Infrequent but still engaged
At Risk	RFM ≥ 6	Purchasing less often — early churn signal
Dormant	RFM < 6	Inactive for an extended period

Customers per segment % of 4,373 total



MAXIMIZE LOYALTY

Champions & Returning — VIP perks, personalised discounts and premium experiences to protect the 28.9% already delivering the most value.

LIFT FREQUENCY

Less Frequent but Loyal — time-sensitive offers and reminder emails to convert occasional engagement into a habit.

RE-ENGAGE & RETAIN

Dormant & At Risk — 47.6% of the base needs win-back campaigns, abandoned-cart emails and direct service follow-ups before it fully churns.

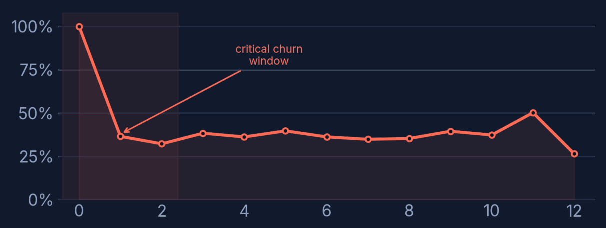
RETENTION

The first two months decide the relationship

Customers were grouped into monthly cohorts by their first purchase date, then tracked for how many stayed active each month after.

December 2010 cohort active customers by month since first purchase

Retention curve % of month-0 base



73.4%

of the December 2010 cohort had disengaged by month 12 (886 → 236 active customers).

77.9%

drop-off in the January 2011 cohort by month 1 alone (417 → 92 active) — the steepest early fall observed.

Month 11

shows a rebound to 446 active customers — a seasonal promotion effect strong enough to temporarily reverse the churn curve.

What's driving early churn

Likely causes: limited post-purchase engagement, unmet quality or service expectations, stronger competitor offers, and no structured incentive to return. The dataset can show **that** customers leave in months 2–3, but not fully **why** — closing that gap needs direct customer feedback layered on top of transaction data.

PRODUCT AFFINITY

A handful of products are quietly selling each other

Market basket analysis mines invoice-level co-purchases to surface which products reliably travel together.

Top 10 frequently bought-together pairs co-occurrence count

Counts reconstructed from the source analysis; all ten pairs exceed 600 joint purchases, led by 22386 + 85099B at 833.

THE ANCHOR PRODUCT

Stock code **85099B** appears in 4 of the top 5 pairs — a single high-demand item worth protecting on stock availability and featuring across every related product page.

COLLECTION BEHAVIOUR

Sequentially-coded pairs (22697/22698/22699, 22726/22727) show customers buying whole product lines — a strong signal to merchandise as curated sets, not single SKUs.

Turning affinity into action

Bundle & upsell

Combo pricing on 85099B + 22386; discount 20725 when bought with 20727; pre-packaged seasonal gift sets from the strongest pairs.

Recommend

Feature paired products on product pages; trigger "customers who bought this also bought..." emails post-purchase.

Stock & place

Keep paired items co-located in the warehouse and stocked ahead of seasonal peaks to protect fulfilment on high-affinity combinations.

From diagnosis to a prioritised action plan

Three analyses, one playbook — each recommendation traceable back to a specific number on the pages before this one.

01

Protect the top of the pyramid

Give Champions and Returning & Loyal customers (28.9% of the base) VIP perks, early access and personalised offers — retention here is far cheaper than acquisition.

02

Win back the 47.6% at risk or dormant

Launch structured win-back campaigns — abandoned-cart triggers, reactivation discounts, and direct service outreach — before the At Risk segment converts to Dormant.

03

Own the first 60 days

Build a structured onboarding sequence for new customers with incentives for a second purchase inside 30–60 days — the window where most cohorts lose over half their base.

04

Bundle around 85099B and its neighbours

Turn the top 10 product pairs into combo pricing, curated sets and on-page recommendations to lift average order value.

05

Fix the December leak

Tighten returns policy and post-purchase communication ahead of peak season — 41.1% revenue loss in December is the single largest recoverable number in this study.

EFFORT

Mostly CRM & lifecycle work

Items 01–03 run through existing email and CRM tooling — no new infrastructure required to start.

MERCHANDISING

Bundle logic is ready today

Item 04 draws directly from the top-10 pairs already surfaced — no further analysis needed before testing offers.

POLICY

Item 05 has the biggest single number

£894K lost to cancellations across the year makes the December returns fix the highest-value single change on this page.

A 90-day sequence, not a 90-page backlog

0–30

Stop the bleeding

- Tighten return & cancellation policy ahead of the next peak season
- Launch onboarding sequence for new customers' first 30 days
- Stand up win-back email flow for the At Risk segment

30–60

Systematise retention

- Deploy RFM-based segmentation into the CRM / email platform
- Launch first bundled offers around top market-basket pairs
- Begin customer feedback collection to diagnose month 2–3 churn drivers

60–90

Scale what works

- Expand seasonal-style promotions (the Nov playbook) into other quarters
- Review cohort curves for the newly onboarded customers
- Formalise VIP tier for Champions and top Returning & Loyal customers

Measuring success

Track three numbers monthly against this baseline: cancellation rate (baseline ~20%, December 37.3%), the size of the At Risk + Dormant segment (currently 47.6%), and month-2 cohort retention. Movement on any of the three is a direct read on whether the playbook is working.

Baseline scorecard

Metric	Current baseline	Direction to watch
Cancellation rate	~20% avg · 37.3% Dec	Down toward the ~18% Q3 low
At Risk + Dormant share	47.6% of customers	Down as win-back campaigns land
Month-2 cohort retention	~35–40% of month-0	Up as onboarding takes effect
Revenue lost to cancellations	£894K / year	Down, concentrated in Q4

CONCLUSION

Key takeaways

RFM segmentation, cohort analysis and market basket analysis converted a flat transaction export into a structured view of who to protect, who to win back, and what to sell together.

Retention & buying behaviour

- 27.1% of customers are at risk of churn — the largest single segment
- Only 7.97% are fully engaged Champions
- Over half of customers stop purchasing within 2–3 months of first buying

Revenue & seasonality

- Revenue mirrors order volume — November (£1.52M) is the clear peak
- February is the weakest month (£0.52M), a post-holiday slump
- December cancellations (41.1% lost revenue) demand better returns management

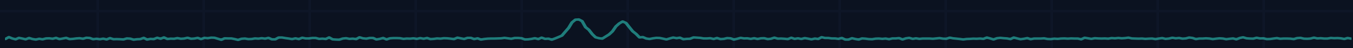
Product & basket insights

- 85099B is the strongest bundling and upsell candidate in the catalogue
- Sequential product codes reveal curated-collection buying behaviour
- 600+ co-occurrence pairs should inform recommendations and inventory

Business implications

- Retention efforts must concentrate on the first 30–60 days
- VIP and loyalty programs should be strengthened for the long term
- Seasonal marketing should extend beyond Q4 to stabilise the year

Moving forward, the highest-leverage next step is closing the "why" gap behind early churn — layering direct customer feedback and behavioural tracking onto this transactional foundation to make retention efforts proactive rather than reactive.



4,373	£10.7M	27.1%	73.4%	833
customers	revenue	at risk	y1 churn	top basket pair

RETAILPULSE

E-Commerce Customer & Revenue Intelligence Platform

Built on the UCI Online Retail dataset · 541,909 transactions · December 2010 – December 2011 · SQL pipeline, Python analysis, Tableau visualisation.

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